

Unneeded faces and missing cullface in dragon egg model

MC-199238 Textures and models

Status	Resolved / Fixed
Affected	1.16.2, 1.16.3, 1.16.4, 20w46a, 20w51a, 21w03a (+34 more)
Fixed in	24w33a
Reported	2020-08-29

STEPS TO REPRODUCE

1. Launch Minecraft 21w41a.
2. Create or load a world.
3. Place a dragon egg.
4. Enter Spectator mode.
5. Fly inside the dragon egg.
6. Observe the stack of cuboids and the faces that are invisible when viewed from outside.
7. Exit Spectator mode.
8. Place an opaque block on top of the dragon egg and another opaque block on the bottom.
9. Enter Spectator mode and fly into the opaque blocks adjacent to the dragon egg.
10. Observe that the dragon egg faces against the opaque blocks remain visible despite being completely occluded.

OBSERVED BEHAVIOUR

Actual behavior

The dragon egg's vanilla block model contains faces for every side of each cuboid that makes up the egg, including faces that are fully enclosed by the model and cannot be seen during normal gameplay. These redundant internal faces are nevertheless submitted for rendering and become visible when the camera clips into the egg, such as in Spectator mode.

The model also omits `cullface` on its uppermost and lowermost faces. Consequently, when an opaque block is placed directly above or below the egg, those faces are still rendered even though the adjacent block completely occludes them.

EXPECTED BEHAVIOUR

When viewing a dragon egg, only faces that can be seen during normal gameplay should be rendered; internal faces exposed only by clipping into the egg should remain invisible. The top and bottom faces should also be properly culled when opaque blocks are placed against them, so completely occluded surfaces do not remain visible from inside those blocks.

ENVIRONMENT

Minecraft Java Edition Versions: 1.16.2, 1.16.3, 1.16.4, 20w46a, 20w51a, 21w03a, 1.16.5, 21w05b, 21w06a, 21w08b, 21w11a, 21w13a, 21w14a, 21w16a, 21w17a, 21w18a, 21w19a, 21w20a, 1.17 Pre-release 1, 1.17, 1.17.1, 21w39a, 21w40a, 21w42a, 21w44a, 1.18, 1.18.1 Pre-release 1, 1.18.1, 1.18.2, 1.19, 1.19.2, 1.19.4 Pre-release 3, 1.19.4, 23w14a, 1.20 Pre-release 1, 1.20.1, 23w43a, 1.21, 1.21.1 Release Candidate 1, 1.21.1 (includes snapshots)

ORIGINAL DESCRIPTION

The resource pack attached offers a complete fix for this issue. Mojang have my full permission to (and are encouraged to) replace the affected vanilla model files with the contents of the resource pack.

Download resource pack:

- 21w41a: dragon-egg-fix-21w41a-v1.1.zip

The bug

The dragon egg includes numerous block faces which cannot be seen normally - in order to see these faces, clipping into the block is required, such as through Spectator mode. In addition, the topmost and bottommost faces are missing cullface specifications standard of other models which have faces which end up hidden by adjacent blocks.

How to reproduce

- Place down a dragon egg
- Enter Spectator mode
- Fly inside of the dragon egg

You will find yourself inside of a stack of cuboids, with many of the faces of these cuboids completely invisible from outside

- Now place opaque blocks on the top and bottom of the egg
- Fly into these opaque blocks

Looking closely reveals that the dragon egg faces against these blocks are still visible despite their complete occlusion

Expected results

Only faces which are visible during normal gameplay are rendered.

Actual results

Every single face of each cuboid, including the ones which are only visible if the player is inside of the dragon egg, is always rendered.

How to fix

The resource pack attached to this ticket fixes this issue completely - all lines pertaining to these redundant faces are deleted (which shrinks the model file considerably), and the top and bottom faces now have cullface specified.

The model also fixes

MC-120417

Resolved

, and re-enables ambient occlusion for the model, as it was disabled before despite offering little to no visual difference.

Formatting has also been changed for consistency and readability.

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-199238>